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Phase 0 — Load to Raw

The raw dataset contains four file types per race: **laps**, **weather**, **session_results**, and **race_control**. A Python load script (**load/load.py**) consolidates all files by:

1. **Parsing filenames** to extract the year, circuit, and file type
 2. **Adding context fields** (**race_year**, **circuit**, **race_id**) to each row
 3. **Loading data** into separate tables (**raw_lap_times**, **raw_weather**, etc.)
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Phase 1 — Staging

Three staging models clean the raw data:

- **stg_lap_times**: Converts **LapTime** to seconds, removes invalid laps, and renames columns for consistency
- **stg_session_results**: Adds **is_finished** flag and standardizes driver IDs to lowercase
- **stg_weather**: Aggregates weather per race with **is_wet** flag and average temperature metrics

Data types were corrected (VARCHAR to seconds for lap times), and tests ensure non-null values for key fields like **driver_id**, **race_id**, and **lap_time_seconds**.

Phase 2 — Intermediate Model

Analytical Questions

Pre-defined question: What is the lap time variability for each driver within each race?

Additional question: How does lap time consistency vary between wet and dry race conditions for each driver, and does this consistency pattern differ by circuit?

Both questions focus on understanding driver consistency, which is critical for race strategy decisions like pit stop timing and tire management. The additional question adds weather as a dimension to help engineers understand how drivers perform under different conditions.

Foundation

Required staging models:

- **stg_lap_times** – provides lap-by-lap timing data
- **stg_session_results** – provides driver finish status
- **stg_weather** – provides weather context per race

Join keys:

- `DriverNumber` (common to both lap times and session results)
- `race_id` (ensures correct race matching)

Base fields carried forward: `driver_id`, `team_name`, `race_id`, `circuit`, `race_year`, `lap_number`, `lap_time_seconds`, `tire_compound`, `tire_life`

Added Fields

Field	Definition	Why Needed
<code>total_laps</code>	COUNT(lap_number) per driver per race	Counts how many laps the driver completed
<code>avg_lap_time</code>	AVG(lap_time_seconds)	Baseline for calculating consistency
<code>lap_time_stddev</code>	STDDEV(lap_time_seconds)	Primary measure of lap time variation
<code>lap_time_range</code>	MAX - MIN lap time	Simple measure of lap time spread
<code>lap_time_cv</code>	(stddev / avg) * 100	Normalized consistency measure for comparing across circuits
<code>is_wet</code>	From weather staging	Enables wet/dry comparison
<code>weather_condition</code>	From weather staging	Human-readable weather label
<code>avg_track_temp</code>	From weather staging	Temperature context for analysis
<code>track_temp_variation</code>	From weather staging	Temperature stability during race

The **Coefficient of Variation (CV)** is the key metric because it normalizes by average lap time, allowing fair comparisons across circuits with different lap lengths.

Intermediate Model Design

The intermediate model (`int_race_driver_consistency`) serves as the analytical foundation for all fact tables. It combines lap times with session results (filtering to finished drivers) and weather data, then groups by driver and race to calculate consistency metrics.

Granularity: One row per driver per race

Key operations:

1. Filter to finished drivers using `stg_session_results`
2. Join lap times using `DriverNumber` and `race_id`
3. Add weather context from `stg_weather`
4. Group by driver and race to calculate statistics

The DBML diagram in `design.dbml` reflects this structure with all base and derived fields.

Phase 3 — Star Model

Analytical Questions

question: What is the lap time variability for each driver within each race? **question:** How does lap time consistency vary between wet and dry race conditions for each driver, and does this consistency pattern differ by circuit?

Each Row Represents (Grain)

Fact Table	Grain
fct_driver_race_consistency	Each row represents one driver in one race
fct_driver_season_consistency	Each row represents one driver in one season
fct_circuit_weather_consistency	Each row represents one circuit with one weather condition in one season

Metrics

fct_driver_race_consistency (Primary Granularity)

Metric name	Source field	Statistical operation	Justification
consistency_stddev	lap_time_seconds	STDDEV	Measures absolute variation in lap times
consistency_cv	lap_time_seconds	(STDDEV / AVG) * 100	Normalized measure for comparing across circuits
consistency_range	lap_time_seconds	MAX - MIN	Simple measure of lap time spread
avg_lap_time	lap_time_seconds	AVG	Baseline average pace
total_laps	lap_number	COUNT	Number of laps completed

fct_driver_season_consistency (Roll-up 1)

Metric name	Source field	Statistical operation	Justification
avg_consistency_cv	consistency_cv	AVG	Overall season consistency
avg_consistency_wet	consistency_cv (wet only)	AVG	Consistency specifically in wet conditions

Metric name	Source field	Statistical operation	Justification
avg_consistency_dry	consistency_cv (dry only)	AVG	Consistency specifically in dry conditions
racess_completed	race_id	COUNT DISTINCT	Number of races analyzed
temp_consistency_correlation	avg_track_temp & consistency_cv	CORR	Relationship between temperature and consistency

fct_circuit_weather_consistency (Roll-up 2)

Metric name	Source field	Statistical operation	Justification
avg_consistency_cv	consistency_cv	AVG	Average consistency for this circuit-weather combo
avg_consistency_stddev	consistency_stddev	AVG	Average variation in lap times
drivers_analyzed	driver_id	COUNT DISTINCT	Number of drivers represented
racess_analyzed	race_id	COUNT DISTINCT	Number of races represented
avg_track_temp	avg_track_temp	AVG	Average temperature for this condition

Star Model Design

The star schema consists of two dimension tables and three fact tables.

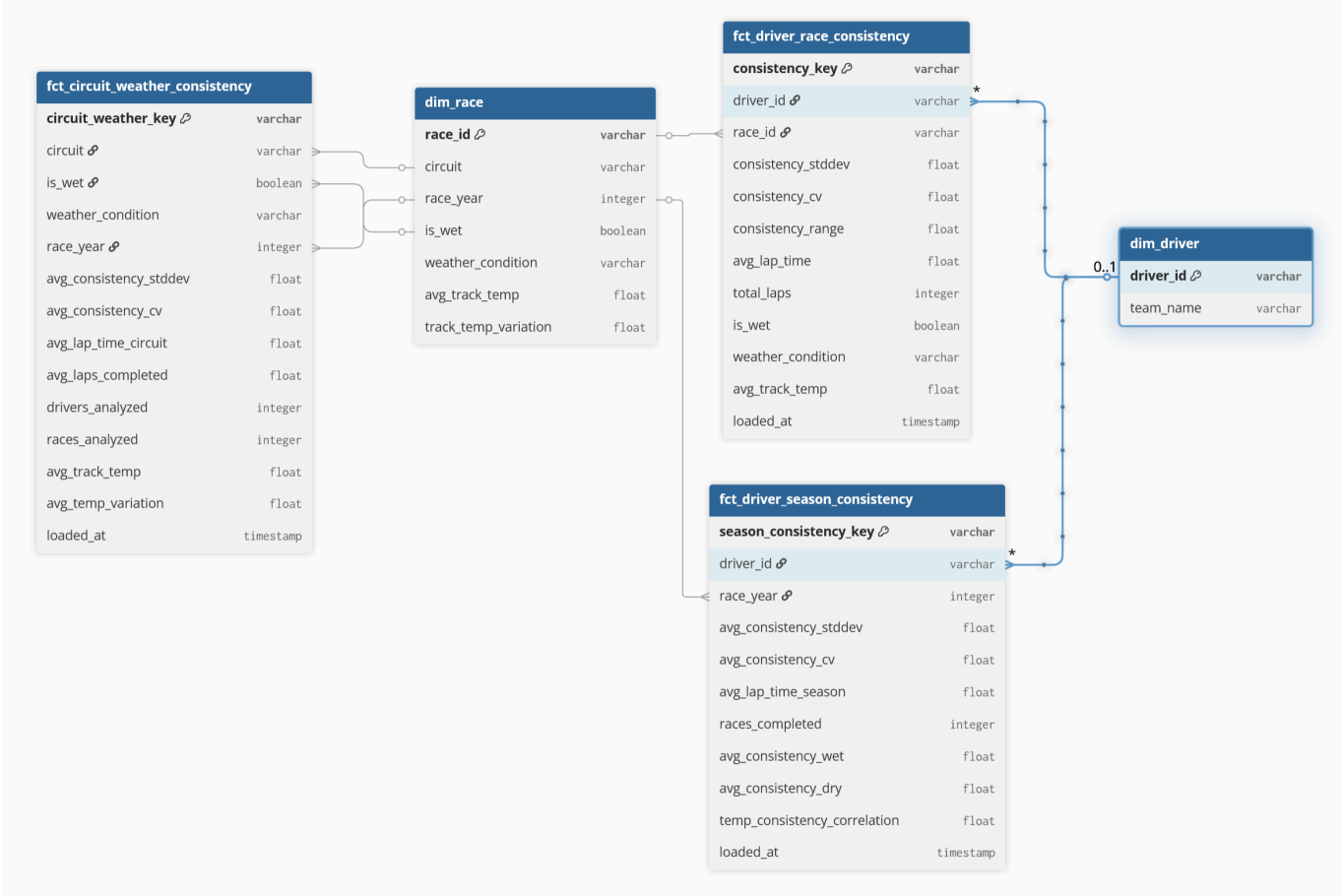
Dimensions:

- dim_driver: Contains unique driver IDs and team names
- dim_race: Contains unique race IDs with circuit, year, and weather context

Fact Tables:

- fct_driver_race_consistency (primary granularity) – links to both dimensions
- fct_driver_season_consistency (roll-up) – links to driver and race dimensions via driver_id and race_year
- fct_circuit_weather_consistency (roll-up) – links to race dimension via circuit, is_wet, and race_year

The design allows analysts to drill down from season-level or circuit-level views to individual driver-race performance. All fact tables share common metrics (CV, stddev) but at different aggregation levels, enabling flexible analysis for race strategy decisions.



int_race_driver_consistency

driver_id	varchar
team_name	varchar
race_id	varchar
circuit	varchar
race_year	integer
total_laps	integer
avg_lap_time	float
lap_time_stddev	float
min_lap_time	float
max_lap_time	float
lap_time_range	float
lap_time_cv	float
is_wet	boolean
weather_condition	varchar
avg_track_temp	float
track_temp_variation	float

Phase 4 — Dashboard



Grain and Data Interpretation

The grain of our data significantly impacts how stakeholders interpret driver consistency patterns. Our three fact tables represent different levels of aggregation, each providing unique insights for decision-making:

- 1. Driver-Race Granularity** (`fct_driver_race_consistency`): This level shows individual driver performance in specific races, enabling teams to evaluate:
 - Driver-specific performance under varying conditions
 - Consistency patterns within single races
 - Pit strategy effectiveness
- 2. Driver-Season Granularity** (`fct_driver_season_consistency`): This roll-up level provides a broader view of driver capabilities across an entire season:
 - Overall consistency trends for drivers
 - Performance in wet vs. dry conditions
 - Seasonal improvement or decline patterns
- 3. Circuit-Weather Granularity** (`fct_circuit_weather_consistency`): This perspective helps understand track and weather-specific challenges:
 - Circuit characteristics that affect consistency
 - Weather impact on overall racing performance
 - Optimal tire strategies for different conditions

Dashboard Structure and Stakeholder Insights

The dashboard is designed to support three primary user groups with distinct analytical needs:

Race Strategy Team

- **Key Metrics:** Consistency CV, average lap time, wet/dry comparison
- **Insights Gained:**
 - Which drivers show the most consistent performance in wet conditions
 - Optimal timing for pit stops based on driver consistency patterns
 - Circuit-specific strategies based on weather impact

Performance Analysts

- **Key Metrics:** Seasonal trends, correlation between temperature and consistency
- **Insights Gained:**
 - Long-term driver performance improvements or declines
 - Relationship between environmental factors and consistency
 - Comparative analysis across different drivers and circuits

Technical Development Team

- **Key Metrics:** Lap time variation, track temperature impact
- **Insights Gained:**
 - Tire compound effectiveness under different conditions
 - Vehicle setup adjustments for maintaining consistency
 - Predictive modeling based on historical performance patterns

The dashboard's drill-down capabilities allow users to move from high-level season trends to specific race-level insights, supporting data-driven decisions at every level of the organization.

Use of Generative AI

I used DeepSeek AI as a technical assistant for this project. The AI was used to help interpret error logs, understand dbt and DuckDB errors, and assist with detailing technical explanations in the report. All code, design decisions, and analytical choices are my own. The AI was used only for debugging support and documentation clarity.